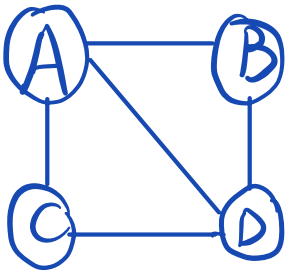
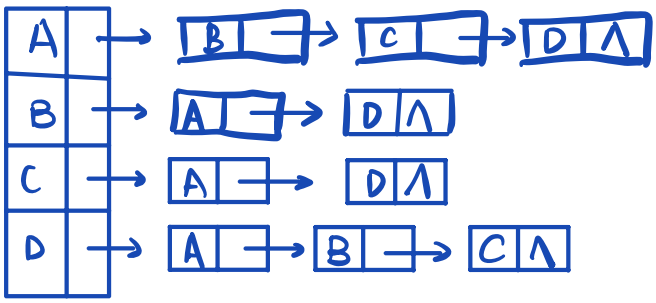
Graph

1. C 2. B 3. C 4. B. 5. D

6. (1) D (2) B 7. A 8. (1) D (2) B

9. (1)

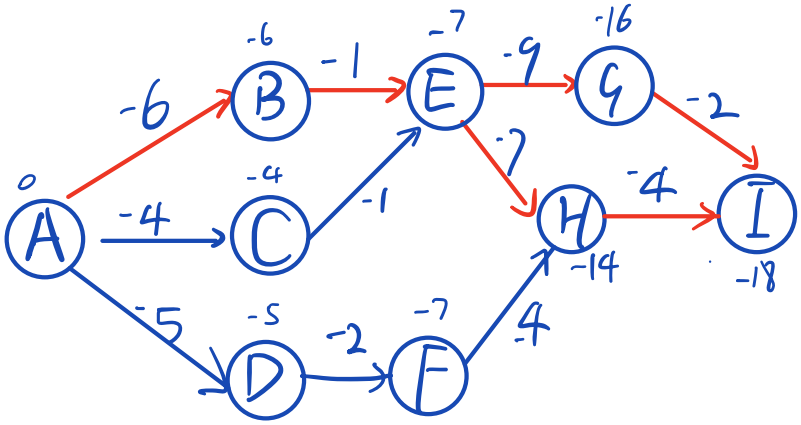




(2) DFS: A B D C

BFS: A B C D

10. Turn into negative:



Time: 18

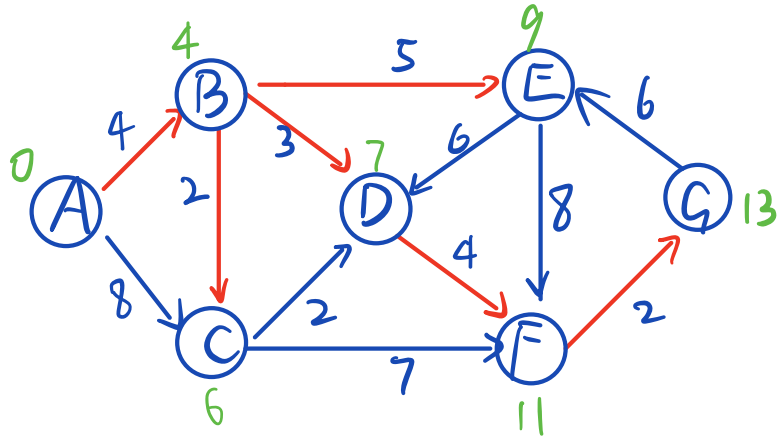
Critical Path:

① A B E G I

② A B E H I

Topological Sort: A D F C B E H G I

11.



B: A → B (4)

C: A → B → C (6)

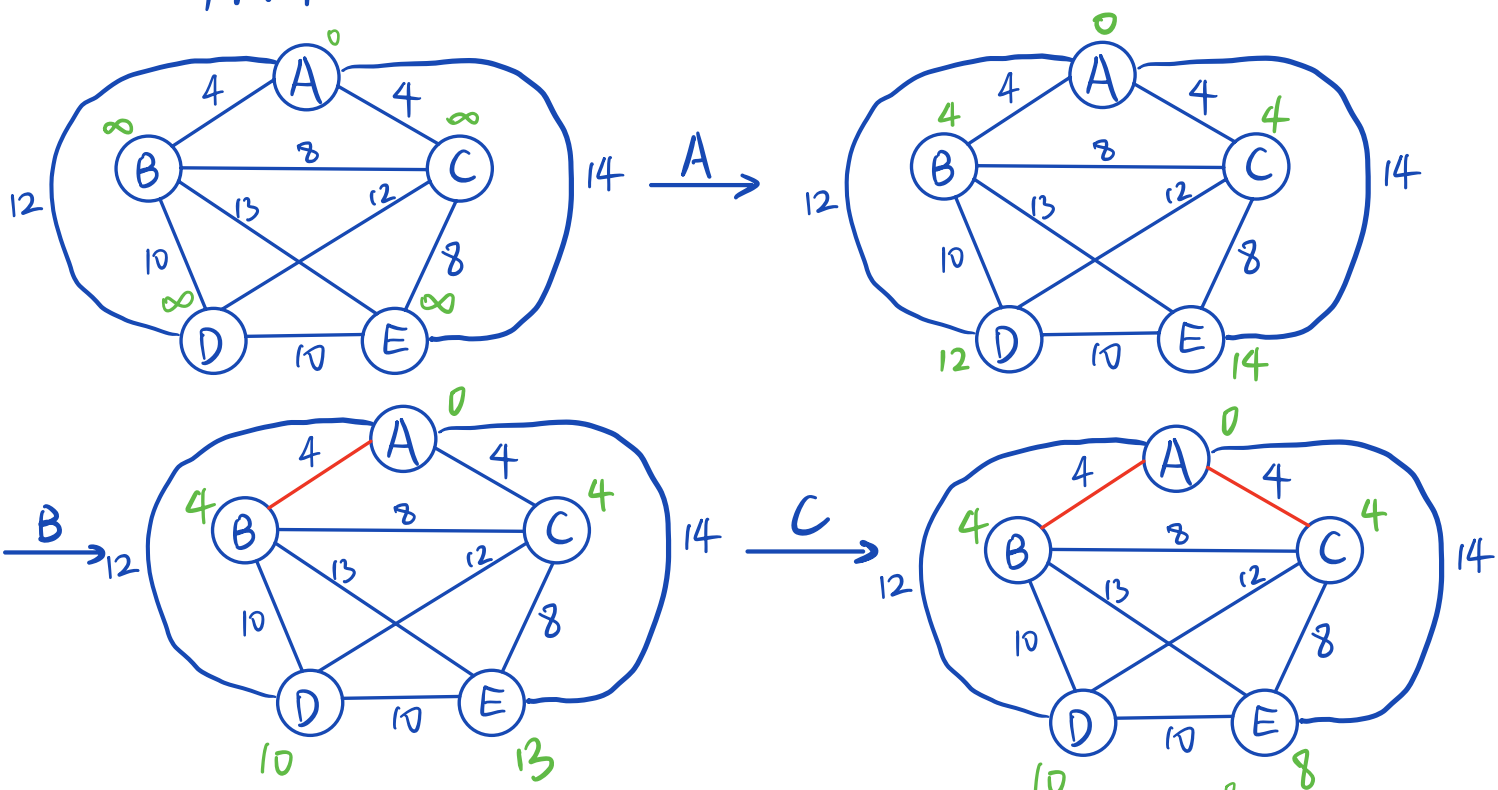
D: A → B → D (7)

E: A → B → E (9)

F: A → B → D → F (11)

G: A → B → D → F → G (13)

12. (1) Prim :



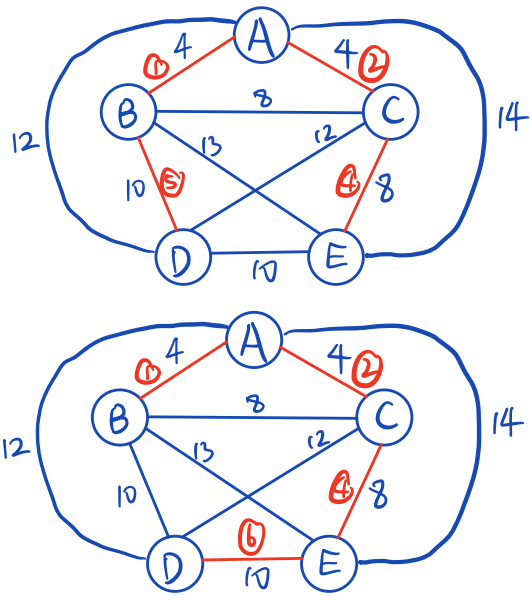
(2) Kruskal :

Sorted edges :

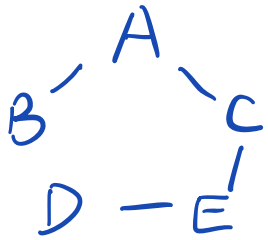
① ~~A-B : 4~~
② ~~A-C : 4~~
③ ~~B-C : 8~~ ✗
④ ~~C-E : 8~~
⑤ ~~B-D : 10~~ ✓
⑥ ~~D-E : 10~~ ✓

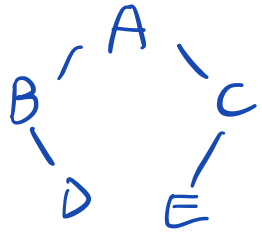
A-D : 12
C-D : 12
B-E : 13
A-E : 14

(Over)



Summary:

(1) MST: ① 

② 

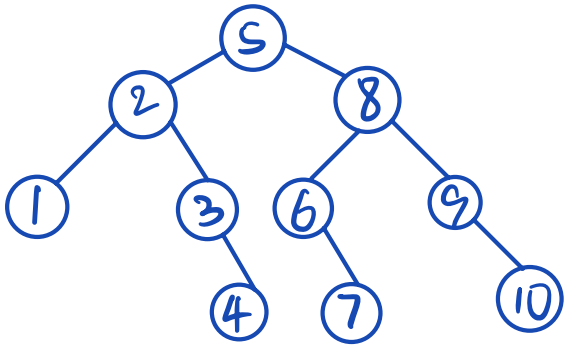
(2) Cost: $4+4+8+10=26$

Search

1. D 2. B.

3. Sorted List: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

BST:



ASL: $\frac{1}{10}(1\times1+2\times2+3\times4+4\times3)=2.9$

4. C. 5. D. 6. D. 7. A.

8. B. $N_1=1$, $N_2=2$

$N_k=N_{k-1}+N_{k-2}+1$

9. D.

10.

Address	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Key			54	94	15			59	47		36	11	89	51	38	63
Order			1	1	3			1	1		1	1	2	2	3	5

ASL $=\frac{1}{11}(1+1+3+1+1+1+1+2+2+3+5)=\frac{21}{11}$

11. Min: $1+2\times2+2\times6=17$

Max: $4+4\times5+4\times25=124$

Sort

1.A 2.D 3.C 4.C 5.A 6.C. 7.C 8.D
9.C 10.B 11.C 12.B

13. 插入排序，归并排序

14. (1) 插入排序

(2) $O(n^2)$

(3) $O(n)$